Supplementary Material for: A Comparative Study of SHAP-Derived and Score-Based Dependency Graphs Using Maximum Weighted Spanning Trees

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ABSTRACT

This document supports the main paper. It contains the comparison of the ground-truth trees with each other and with the ALARM network (Section S1), the performance of the per-variable XGBoost models (Section S2), the edge-type counts and paired tests of the expert-network environment (Section S3), and example SHAP trees learned from ALARM (Section S4). Environment names, methods and metrics are as defined in the main paper. All graph figures use the same fixed layout of the ALARM network, so they can be compared directly.

S1. Ground-Truth Trees of the Pairwise-Association Environment

The ten ground-truth trees were built from pairwise mutual information on forward samples of ALARM of size 1,000 to 10,000 (GT1 to GT10). Figure S1 shows how different they are from one another. Most pairs differ by between 2 and 9 edges, GT1 and GT3 are identical, and so are GT2 and GT10. The trees are therefore similar but not interchangeable, with 36 edges each.

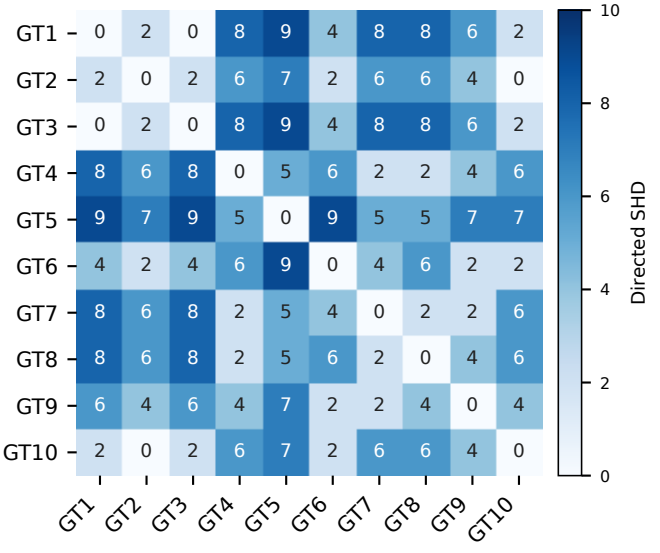


Figure S1: Directed structural Hamming distance between every pair of ground-truth trees. Lighter cells mean more similar trees; the diagonal is zero.

Figures S2 and S3 compare two of these trees with the original ALARM network. Although both give exactly the same summary counts against ALARM (12 correct, 19 reversed, 5 added and 15 missing edges), they are not the same tree: they differ by 8 edges (Figure S1), for example around ERRCAUTER, HREKG and HRSAT, and around

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KINKEDTUBE and PRESS. Their orientation is fixed by the shared root HREKG, which is why so many edges appear reversed relative to ALARM.

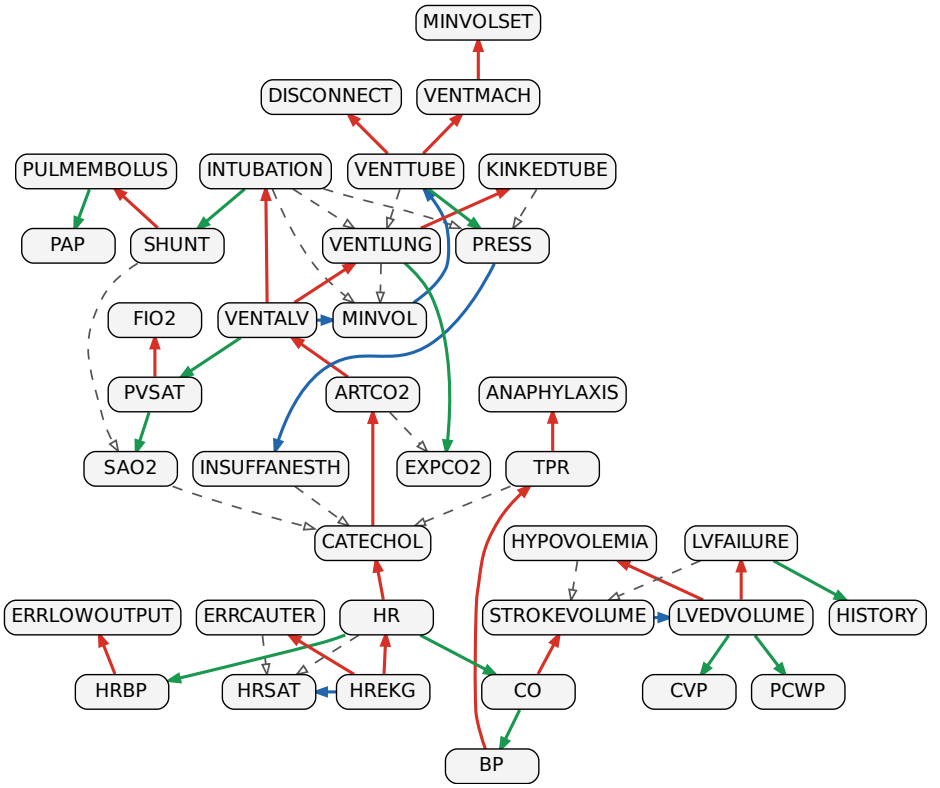


Figure S2: Ground truth GT1 (MI tree from 1,000 ALARM samples) compared with the ALARM network: 12 correct, 19 reversed, 5 added and 15 missing edges. Edge colours: — correct (same edge and direction as ALARM), — reversed, — added (not in ALARM), - - - missing ALARM edge (drawn in ALARM's direction).

S2. XGBoost Model Performance

For every dataset in the pairwise-association environment, one XGBoost classifier was trained per variable (37 models), using 60% of the rows for training, 20% for validation and 20% as a held-out test set. All scores below are measured on the held-out test set, which the model never saw during training. Table S1 reports the test accuracy and weighted F1 score, first averaged over the 37 models of a dataset and then over the nine ground truths with SHAP results (GT5 has none).

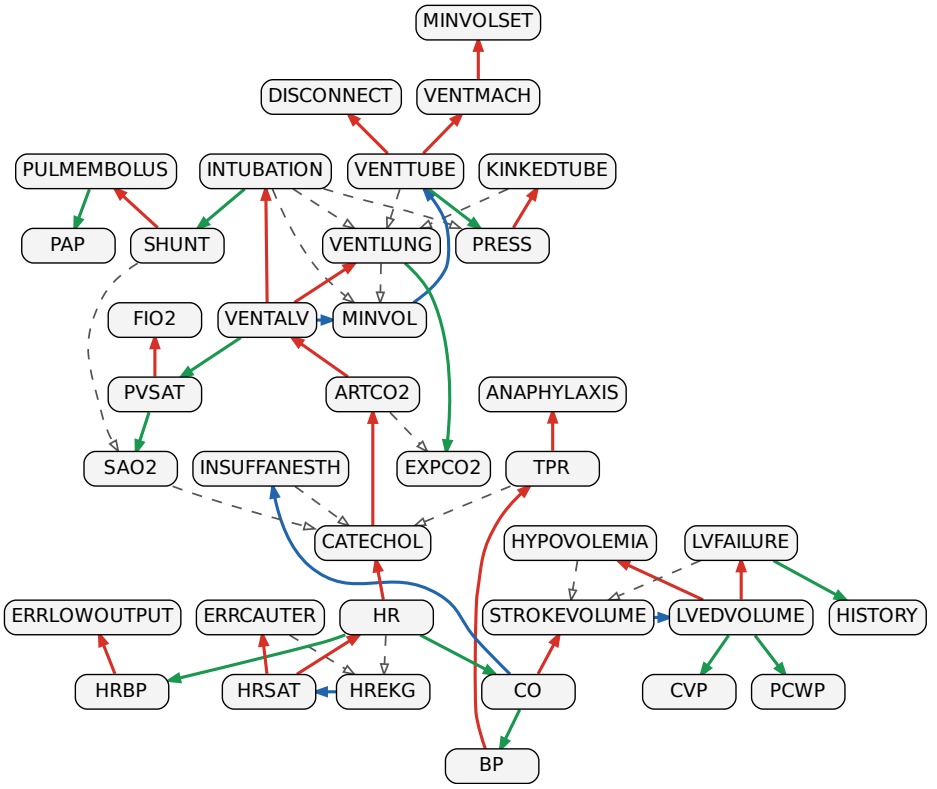


Figure S3: Ground truth GT8 (MI tree from 8,000 ALARM samples) compared with the ALARM network: 12 correct, 19 reversed, 5 added and 15 missing edges, the same totals as GT1 but with different edges. Edge colours: ▬ correct (same edge and direction as ALARM), ▬ reversed, ▬ added (not in ALARM), ▬ missing ALARM edge (drawn in ALARM's direction).

Table S1
Test performance of the per-variable XGBoost models in the pairwise-association environment: mean over 37 models, then mean and standard deviation (SD) across the nine ground truths.

	Accuracy		Weighted F1	
n	Mean	SD	Mean	SD
50	0.597	0.021	0.570	0.016
100	0.606	0.015	0.583	0.017
500	0.656	0.010	0.643	0.010
1000	0.662	0.008	0.648	0.007
2000	0.670	0.009	0.657	0.009
3000	0.673	0.010	0.660	0.010
5000	0.677	0.008	0.664	0.008
6000	0.679	0.008	0.666	0.009

S2.1. Are the Models Acceptable?

The XGBoost models are not the object of study; they are the instruments from which the SHAP weights are read. A model is therefore acceptable if it has captured the real dependence between its target and the other variables, so that its explanations reflect that dependence, rather than if it reaches a particular accuracy. Figure S4 shows the three features of the results that support this.

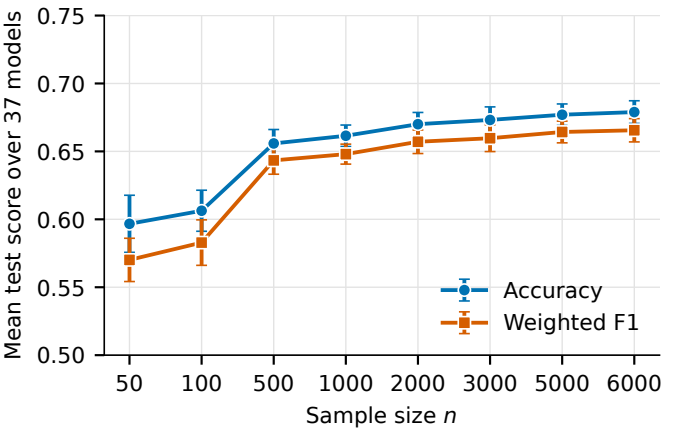


Figure S4: Mean held-out test accuracy and weighted F1 of the 37 XGBoost models against sample size (mean ± one SD across ground truths).

- *The models learn from the data.* Test accuracy rises from 0.60 at $n = 50$ to 0.66 at $n = 500$ and keeps improving slowly after that. A model that had captured no real dependence would not improve as it is given more data from the same distribution.
- *The models generalise.* All scores are measured on test rows that the models never saw. An overfitted model memorises its training rows and scores poorly on new ones; here test performance instead levels off from about $n = 1000$, changing by only 0.01 to 0.02 between $n = 1000$ and $n = 6000$ for every ground truth. There is therefore no sign of harmful overfitting from moderate sample sizes onwards.
- *The behaviour is consistent.* From $n = 500$ the standard deviation across ground truths is below 0.01, so no ground truth produced an unusually poor set of models.

Two qualifications apply. First, at $n = 50$ and $n = 100$ each model trains on only 30 or 60 rows with up to 200 trees of depth 6, so overfitting is likely there, and each test set has only 10 or 20 rows, so the scores are coarse. This matches the large small-sample deficit of the SHAP trees reported in the main paper. Second, training accuracy was not recorded, so the gap between training and test performance cannot be reported directly. The moderate absolute accuracies reflect the ground truths rather than the models: their conditional probability tables were drawn at random from a Dirichlet distribution, so many variables are only weakly predictable from the others even with unlimited data, and Figure S5 shows that predictability differs strongly between variables.

On this evidence the models are a fair basis for the SHAP weights from $n = 500$ onwards, and results at $n = 50$ and $n = 100$ should be read with the limitations above. The structural results of the main paper support the same conclusion: SHAP trees built from these models recover almost the entire skeleton of the ground truth at large sample sizes, which would not be possible if the models had failed to capture the dependence structure.

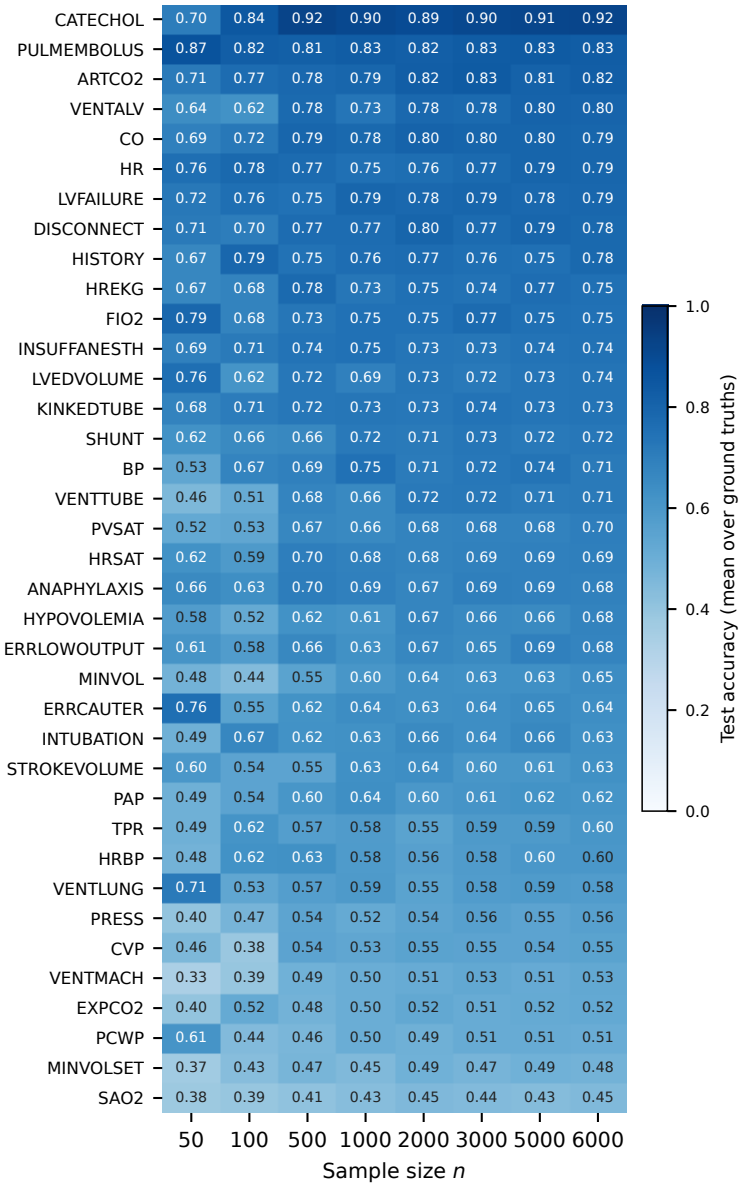


Figure S5: Held-out test accuracy of each target model by sample size, averaged across ground truths. Rows are sorted by accuracy at $n = 6000$, which ranges from 0.45 (SAO2) to 0.92 (CATECHOL).

S3. Expert-Network Environment: Edge Types and Paired Tests

Table S2 breaks the directed SHD of each learner against the ALARM DAG into its four edge types. Tables S3 and S4 give the paired comparisons that underlie the significance statements in the main paper.

Table S2
Expert-network environment: mean edge-type counts against the ALARM DAG over 10 seeds. C = correct (same direction), R = reversed, A = added, D = deleted. Directed SHD = R + A + D. For the two 36-edge trees, D = A + 10 by construction.

	BIC tree				SHAP tree				BIC-HC			
n	C	R	A	D	C	R	A	D	C	R	A	D
50	11.7	8.4	15.9	25.9	7.4	5.8	22.8	32.8	8.3	10.8	9.4	26.9
100	15.2	10.2	10.6	20.6	11.2	7.4	17.4	27.4	9.7	13.3	7.6	23.0
500	15.1	15.3	5.6	15.6	17.6	9.1	9.3	19.3	18.4	14.0	6.7	13.6
1000	16.5	14.5	5.0	15.0	21.8	8.8	5.4	15.4	21.1	17.3	7.5	7.6
2000	16.6	14.3	5.1	15.1	25.2	7.6	3.2	13.2	23.3	16.2	8.0	6.5
3000	17.6	13.4	5.0	15.0	26.5	6.8	2.7	12.7	25.4	15.2	7.3	5.4
5000	14.0	16.9	5.1	15.1	26.7	6.4	2.9	12.9	24.5	16.6	8.7	4.9
6000	15.9	15.1	5.0	15.0	27.1	6.2	2.7	12.7	28.1	13.4	9.0	4.5

Table S3
Expert-network environment: paired comparison, SHAP tree minus BIC tree, over the same 10 seeds. Δ is the mean per-seed difference (negative favours SHAP); "wins" is the number of seeds out of 10 in which SHAP is lower; p is a two-sided Wilcoxon signed-rank test.

	KL		Directed SHD		Skeleton SHD	
n	Δ (wins)	p	Δ (wins)	p	Δ (wins)	p
50	+0.438 (4)	0.193	+11.2 (0)	0.004	+13.8 (0)	0.002
100	+0.476 (3)	0.193	+10.8 (0)	0.002	+13.6 (0)	0.002
500	+0.722 (1)	0.006	+1.2 (2)	0.242	+7.4 (0)	0.002
1000	+1.172 (1)	0.004	−4.9 (9)	0.027	+0.8 (3)	0.539
2000	+0.581 (1)	0.027	−10.5 (10)	0.002	−3.8 (9)	0.004
3000	−0.145 (4)	0.695	−11.2 (10)	0.002	−4.6 (10)	0.002
5000	−0.196 (8)	0.193	−14.9 (10)	0.002	−4.4 (9)	0.004
6000	−0.169 (8)	0.131	−13.5 (10)	0.002	−4.6 (10)	0.002

Table S4
Expert-network environment: paired comparison, SHAP tree minus BIC hill climbing, over the same 10 seeds. Notation as in Table S3.

	KL		Directed SHD		Skeleton SHD	
n	Δ (wins)	p	Δ (wins)	p	Δ (wins)	p
50	+2.274 (0)	0.002	+14.3 (0)	0.002	+19.3 (0)	0.002
100	+2.469 (0)	0.002	+8.3 (0)	0.002	+14.2 (0)	0.002
500	+3.133 (0)	0.002	+3.4 (1)	0.047	+8.3 (0)	0.002
1000	+3.304 (0)	0.002	−2.8 (7)	0.072	+5.7 (1)	0.004
2000	+3.256 (0)	0.002	−6.7 (9)	0.004	+1.9 (2)	0.125
3000	+3.080 (0)	0.002	−5.7 (8)	0.008	+2.7 (1)	0.012
5000	+3.061 (0)	0.002	−8.0 (10)	0.002	+2.2 (2)	0.021
6000	+3.034 (0)	0.002	−5.3 (8)	0.021	+1.9 (2)	0.094

S4. SHAP Trees Learned from the ALARM Network

Figures S6 and S7 show the SHAP tree learned from forward samples of ALARM for seed 42 at the smallest and the largest sample size, compared edge by edge with the ALARM network. With 50 observations the tree is dominated by added and missing edges. With 6,000 observations most of its edges are correct and correctly directed, illustrating both the structural recovery and the orientation accuracy reported in the main paper. The 12 missing edges at $n = 6000$ include the 10 that any 36-edge tree must leave out.

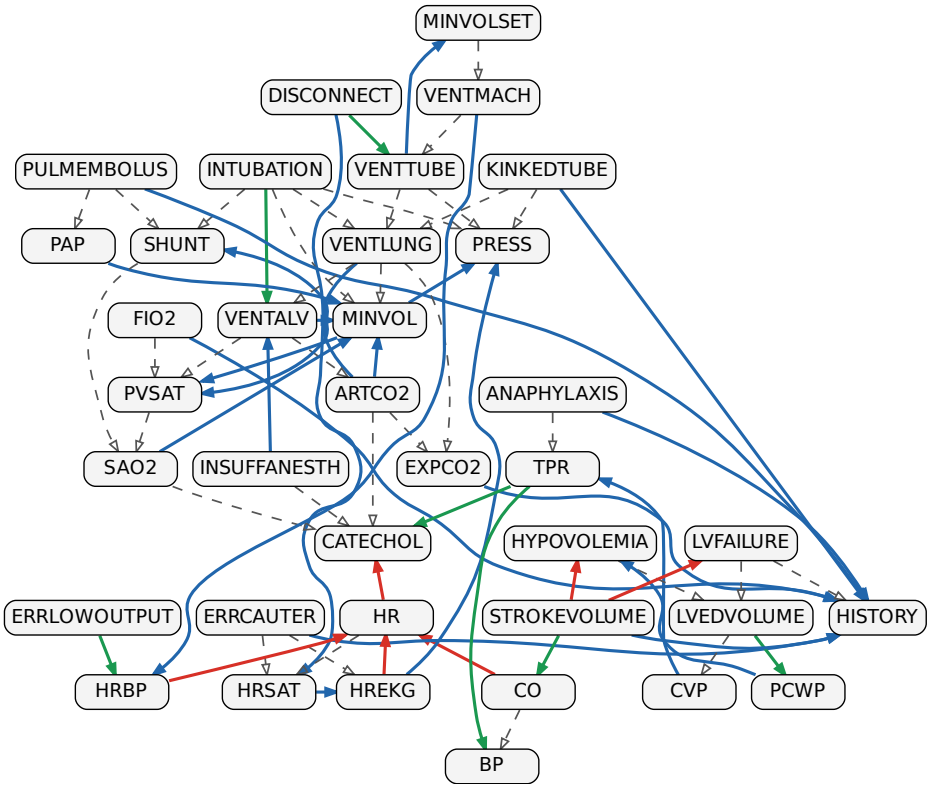


Figure S6: SHAP tree learned from 50 ALARM samples (seed 42) compared with the ALARM network: 7 correct, 6 reversed, 23 added and 33 missing edges. Edge colours: — correct (same edge and direction as ALARM), — reversed, — added (not in ALARM), - - - missing ALARM edge (drawn in ALARM's direction).

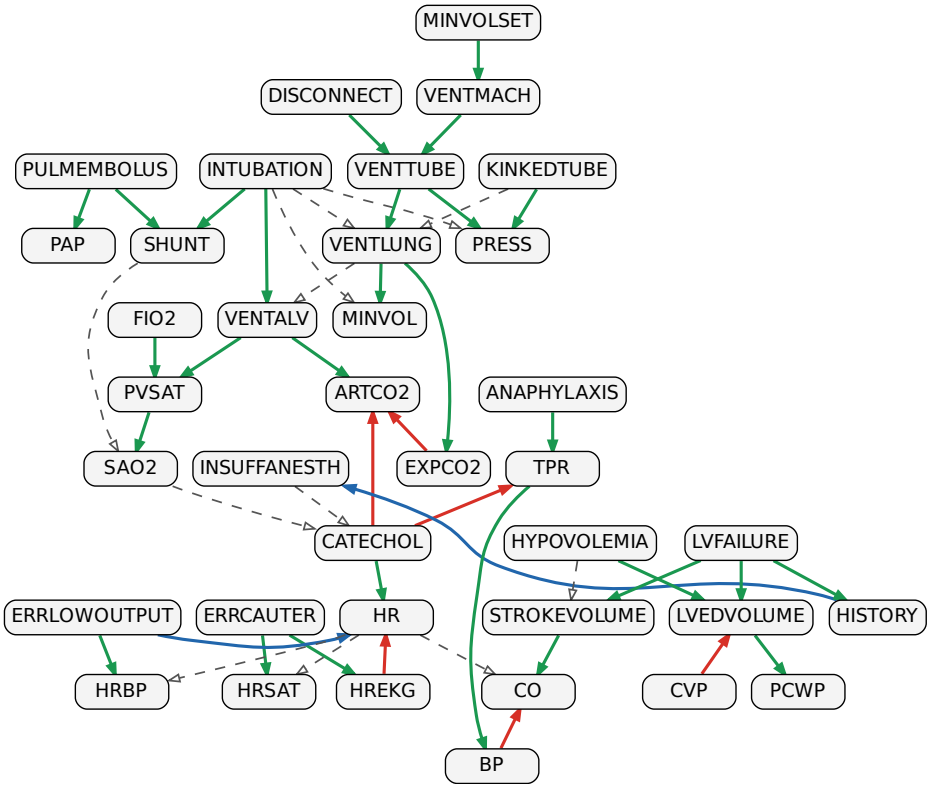


Figure S7: SHAP tree learned from 6,000 ALARM samples (seed 42) compared with the ALARM network: 28 correct, 6 reversed, 2 added and 12 missing edges. Edge colours: ——— correct (same edge and direction as ALARM), ——— reversed, ——— added (not in ALARM), - - - missing ALARM edge (drawn in ALARM's direction).